

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Microprocessor

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- 1) Enlist the greatest breakthrough in microprocessor so that modern processors are available for personal computer. 5
- 2) With reference of bus advancement differentiate between Harvard Architecture and Von Neumann architecture. 5
- 3) Explain with example the addressing mode of 8085 microprocessor. 5
- 4) Define instruction set and formats, opcodes, mnemonics and operands with examples. 8
- 5) Write an 8085 program for the following type of addition 7
$$1^2 + 2^2 + 3^2 + \dots + 9^2$$
- 6) Define T-state. Draw the timing diagrams for the instruction OUT 0FH. 8
- 7) Write a program to take input from 4- switches connected to PC3-PC0 and display the status of the switches to 4 LEDs connected to PC7 -- PC4 of 8255 PPI. 7
- 8) a) What is memory interfacing? Interface two PROM and one RAM with 8KB and 4KB memory respectively with 8085. 8
- b) Explain the Programmable Interrupt Controller (8259 A) with a suitable block diagram. 7
- 9) a) Do you think micro assembler is essential for microprocessor to execute the program? If yes, explain with the difference between
 - i. END and ENDP
 - ii. DB and DW
 - iii. PROC and EXTRN
 - iv. PUBLIC and GLOBAL7
- b) Write an 8086 program to find square root of a given number. Given 8

- that number is a perfect square of two digits.
6. a) Explain with example the flags of 8086 microprocessor.
 - b) Write the difference between serial and parallel communication. Explain the RS232C standard for serial communication.
 7. Write short notes on: (Any two)
 - a) 8255 PPI
 - b) 80386 microprocessor and its type
 - c) Procedures and Macros